
Phase 1, Module 1: Core Syntax & Professional Environments

Khyber AI Systems Engineer (KASE)

Welcome to KASE. Today, we don't just learn to code; we learn to build an environment where our code can survive in the real world.

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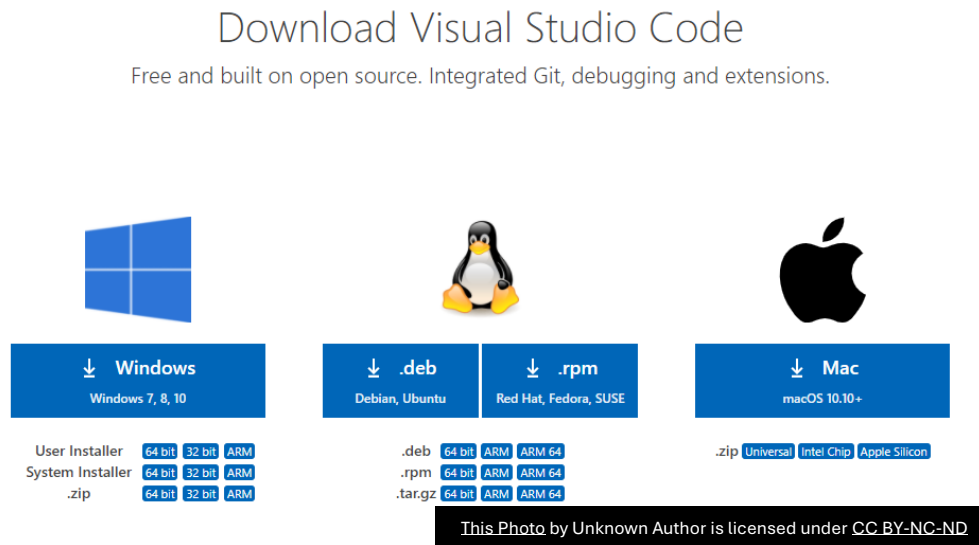
Installing 'The Brain' (Python)



Getting the Python Engine

- Download Python 3.10+ from python.org/downloads.
- CRITICAL: Check the box that says "Add Python.exe to PATH" during installation.
- Without "PATH", your computer will not know Python exists.

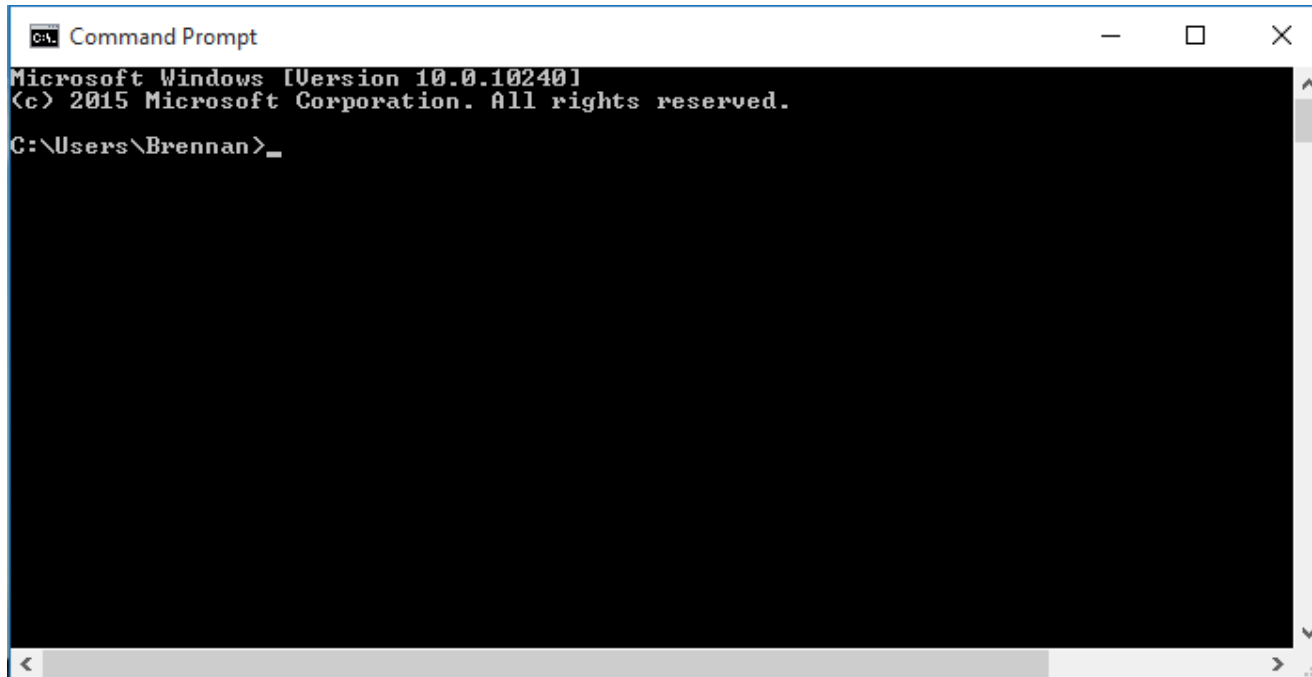
Building the Workbench (VS Code)



Visual Studio Code

- We do not use browser-based Jupyter or basic text editors.
- Download from code.visualstudio.com.
- Install the Python and Jupyter extensions (by Microsoft) inside VS Code.

Verifying the Environment (CLI Test)

A screenshot of a Windows Command Prompt window. The title bar reads "C:\ Command Prompt". The window content shows the following text: "Microsoft Windows [Version 10.0.10240]
<c> 2015 Microsoft Corporation. All rights reserved.
C:\Users\Brennan>_". The prompt is at the end of the third line, followed by an underscore. The window has standard Windows window controls (minimize, maximize, close) in the top right corner and a scrollbar on the right side.

```
C:\ Command Prompt
Microsoft Windows [Version 10.0.10240]
<c> 2015 Microsoft Corporation. All rights reserved.
C:\Users\Brennan>_
```

Talking to Your Computer

- Open your Terminal or Command Prompt.
- Run: `python --version` (Should output Python 3.1x.x).
- Run: `pip --version` (Checks the package installer).

The Python Interpreter

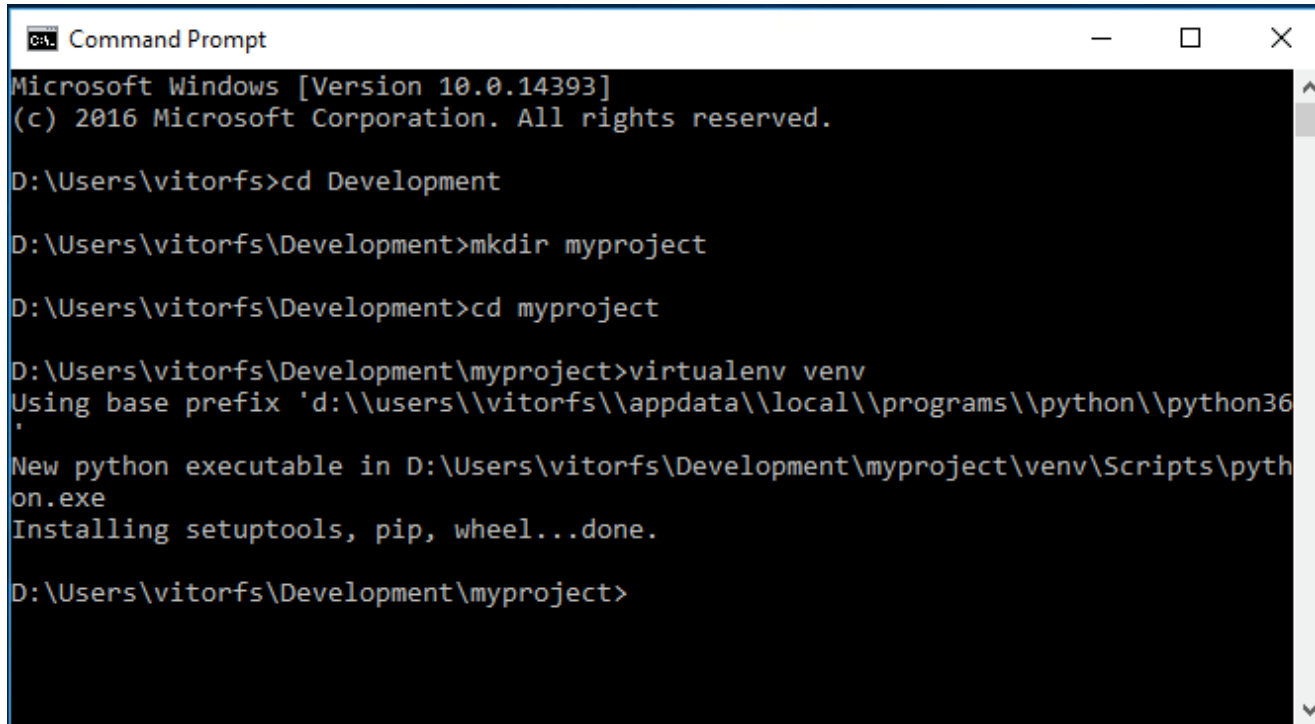
```
1 # Add your Python code here. E.g.  
2 from microbit import *  
3  
4  
5
```

Using snippets is easy. Just follow these instructions...

How Python "Thinks"

- Python is an Interpreted language (reads line-by-line).
- Slower than C++ for the machine, but 10x faster for the human developer.
- The engine behind the global AI revolution.

The Virtual Environment (VENV)



```
Command Prompt
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

D:\Users\vitorfs>cd Development

D:\Users\vitorfs\Development>mkdir myproject

D:\Users\vitorfs\Development>cd myproject

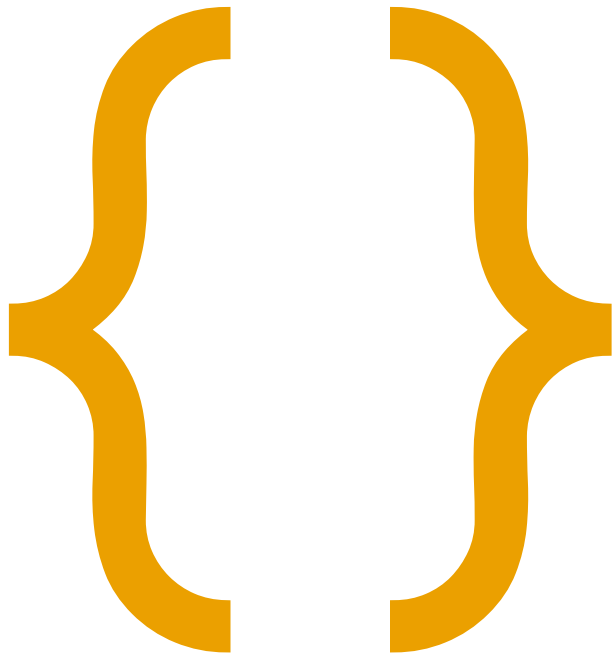
D:\Users\vitorfs\Development\myproject>virtualenv venv
Using base prefix 'd:\\users\\vitorfs\\appdata\\local\\programs\\python\\python36'
New python executable in D:\Users\vitorfs\Development\myproject\venv\Scripts\python.exe
Installing setuptools, pip, wheel...done.

D:\Users\vitorfs\Development\myproject>
```

The Golden Rule: Never Install Globally

- What is it? A quarantined room for your project's tools.
- Why? Prevents version crashes between different projects.
- The Command: `python -m venv kase_env`

PEP 8 - The Law of Clean Code



Writing Code for Humans

- Code is read 100 times more often than it is written.
- PEP 8 is the global style guide.
- Variables: snake_case | Constants: UPPER_CASE.
- Indentation: Exactly 4 spaces.